# AI Sales & Demand Prediction System

*AI Solution - DOCUMENTATION ASPECT*

## 1. AI Solution: Relevance to Theme (Background) [5 marks]

Fuel filling stations across South Africa increasingly operate hybrid retail environments — combining a forecourt with a convenience store and, in many cases, an in-house food kitchen — yet most still rely on manual stock counts and manager intuition to decide what to order and when. This directly reflects the Theme, “an AI Solution for Industries”: while 4IR technologies are widely discussed, small and medium retail operators such as filling stations remain largely excluded from their benefits. This project proposes an AI-powered Sales and Demand Prediction System, built in Python, that treats the store and the kitchen as one integrated system with two linked data streams. The system uses historical sales data (product sold, quantity, price, day of week, promotions, weather conditions and current stock levels) to forecast future demand for both convenience-store products and kitchen ingredients, alerting managers when predicted demand is likely to exceed available stock so that ordering decisions become proactive rather than reactive.

## 2. Business Objectives [25 marks]

### 2.1 Business Background

The client environment is a fuel filling station operating a hybrid retail model: a forecourt, an attached convenience store, and an in-house food kitchen. Ordering decisions for both the store and the kitchen are currently made manually, based on staff experience and periodic stock counts rather than historical sales patterns. This is typical of small and medium filling-station operators in South Africa, who generally lack the budget or technical staff to implement enterprise-grade demand-forecasting systems, despite operating in an industry where demand is highly variable and perishable stock is common.

### 2.2 Business Objectives

- Reduce stockouts of fast-moving convenience and kitchen items by forecasting demand ahead of high-demand periods (weekends, holidays, promotions).
- Reduce waste of slow-moving and perishable kitchen ingredients by aligning order quantities with predicted, rather than assumed, demand.
- Replace manual, experience-based ordering with a data-driven decision-support process for store and kitchen managers.
- Provide early, automated alerts when predicted demand is likely to exceed current stock levels, enabling proactive rather than reactive ordering.
- Improve overall profitability, customer satisfaction and product availability at the filling station.

### 2.3 Business Success Criteria

- At least a 30% reduction in stockout incidents for the top 20 fast-moving store and kitchen items within 6 months of go-live.
- At least a 25% reduction in measurable kitchen ingredient wastage (by value or weight) within 6 months of go-live.
- Demand forecast error (e.g. Mean Absolute Percentage Error) below 15–20% for key product categories.
- At least 80% of restocking decisions made by managers reference the system's alerts or forecasts within 3 months of adoption.
- Measurable reduction (e.g. 50%) in manager time spent on manual stock counting and estimation.

### 2.4 Requirements

- Functional: ingest historical sales, price, promotion, weather and stock-level data; generate short-term demand forecasts per product/ingredient; compare forecasts against current stock and trigger restock alerts; present results via a simple dashboard/report for managers.
- Data: a minimum of 12 months of historical sales data, current stock levels, a promotions calendar, and weather data (via a weather API) for the store's location.
- Technical: Python-based forecasting pipeline (e.g. pandas, scikit-learn, Prophet/statsmodels), a lightweight database for storing sales and stock data,

and a simple front end (e.g. dashboard or Streamlit app) usable by non-technical staff.

- Usability: alerts and forecasts must be understandable to store/kitchen managers without a data-science background.

## 2.5 Constraints

- Limited quality and completeness of existing historical sales data, since records have been kept manually.
- Limited IT budget and infrastructure typical of a small/medium filling-station operator, with no dedicated in-house IT or data-science staff.
- Fixed academic project timeline and scope, limiting the depth of model tuning and integration possible before the due date.
- Dependence on an external weather API and its availability, accuracy and pricing.

## 2.6 Risks

- Forecast accuracy may be low if historical data is sparse, inconsistent or poorly labelled — mitigated by data cleaning and starting with simpler, explainable models.
- Managers may resist adopting a new system and continue ordering manually — mitigated by a simple interface, short training, and phased rollout alongside the existing process.
- A single model may not generalise well across different stores, regions or seasons with different demand patterns — mitigated by per-store/per-category retraining.
- Forecast quality will degrade over time if sales and stock data are not captured consistently — mitigated by lightweight data-entry validation and regular monitoring.
- Data privacy/security risk where the system connects to point-of-sale data — mitigated by restricting access and anonymising data used for testing.

## 2.7 Initial Assessment of Tools and Techniques

- Language: Python, using pandas and NumPy for data preparation and cleaning.
- Forecasting/Machine Learning: regression models (scikit-learn) and time-series methods (ARIMA/SARIMA via statsmodels, or Facebook Prophet) to capture seasonality such as day-of-week and promotional effects.
- Visualisation and reporting: Matplotlib/Seaborn for exploratory analysis, and a simple dashboard (e.g. Streamlit or Power BI) for manager-facing alerts and reports.
- Alerting logic: rule-based comparison of predicted demand against current stock levels, triggering a dashboard notification, email or SMS alert when a shortfall is predicted.
- Optional conversational layer: a lightweight chatbot/softbot allowing a manager to ask, for example, “what is the predicted demand for bottled water this weekend?” using simple intent matching or a small NLP library.

## 3. Problem Definition [10 marks]

Filling station convenience stores and their in-house kitchens currently order stock based on manual estimation and staff experience rather than data-driven forecasting. This causes two recurring problems: stockouts, where fast-moving items such as bottled water, snacks and key kitchen ingredients run out during high-demand periods such as weekends, holidays or promotions, resulting in lost sales and frustrated customers; and overstocking, where slow-moving or perishable items — particularly fresh kitchen ingredients — are ordered in excess and go to waste before they can be sold. Demand fluctuates with variables such as day of the week, weather conditions and promotional activity, making it difficult for staff to predict it accurately across multiple product categories in the store and kitchen at the same time. As a result, stores experience inconsistent stock availability, avoidable financial losses from waste and lost sales, and an inability to make timely, informed purchasing decisions, which undermines profitability and customer service.

This problem is directly relevant to the Theme, “an AI Solution for Industries,” because it applies 4IR technology to a small/medium retail industry currently excluded from most AI adoption. Solving it benefits the business through improved profitability and reduced waste, and benefits the surrounding community — particularly where the

station serves a small town — through more reliable access to essential convenience goods and food, and continued local employment.

## 4. Poster Overview [10 marks]

